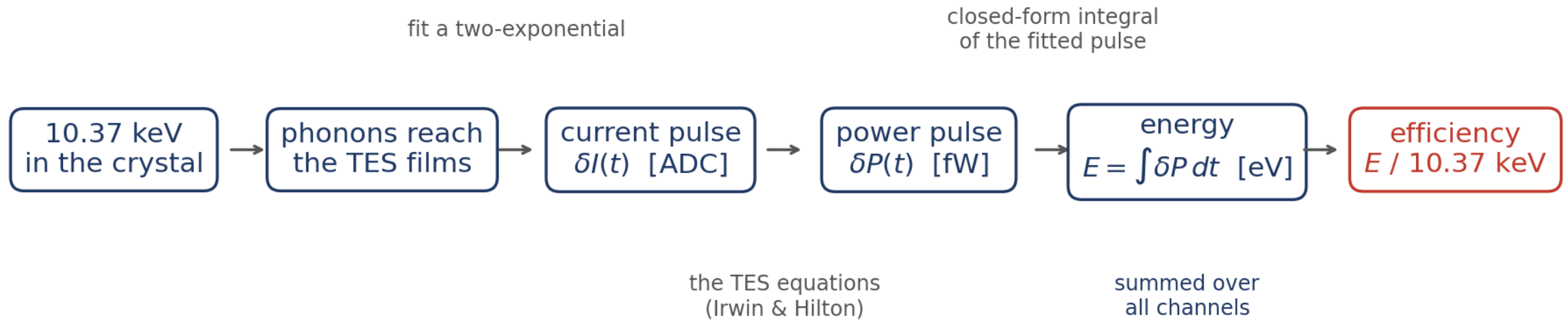


Phonon collection efficiency of Z7

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github.com/ZhihengLi0/urop → [snolab/collection_efficiency/](#)

The Main Idea



For detectors operated at 0 V (no NTL effect):

Collection efficiency = energy absorbed by the TESs, summed over all channels $\div$ the 10.37 keV of the event

Sample: Z7, the K-line events of the Ge activation run, from SNOLAB R4 (2207 events, 30 series, all 11 channels).

The formula, and the two numbers it needs

Method 1 of the CDMS collection-efficiency note

$$\delta P(t) = I_0 (R_L - R_0) \delta I(t) + 2 R_L (\delta I(t))^2$$

linear term

quadratic term

$$E = \int \delta P(t) dt = I_0 (R_L - R_0) A (\tau_f - \tau_r) + 2 R_L A^2 \left(\frac{\tau_f + \tau_r}{2} - \frac{2 \tau_f \tau_r}{\tau_f + \tau_r} \right)$$

for the fitted pulse $\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$ integrated from $-\infty$ to $+\infty$: zero before it starts, decays to zero after

$$\text{Z7 PBS1: } I_0 = -18.07 \mu\text{A}, \quad R_0 = 44.16 \text{ m}\Omega, \quad R_L = R_p + R_{sh} = 14.24 \text{ m}\Omega + 5.00 \text{ m}\Omega = 19.24 \text{ m}\Omega$$

$$\text{linear coefficient: } I_0 (R_L - R_0) = (-18.07 \mu\text{A}) \times (19.24 \text{ m}\Omega - 44.16 \text{ m}\Omega) = 0.45 \mu\text{V}$$

$$\text{quadratic coefficient: } 2 R_L = 2 \times 19.24 \text{ m}\Omega = 38.5 \text{ m}\Omega = 0.0385 \Omega$$

δP = the change of Joule heating when the current moves by δI : a linear term and a small quadratic term.

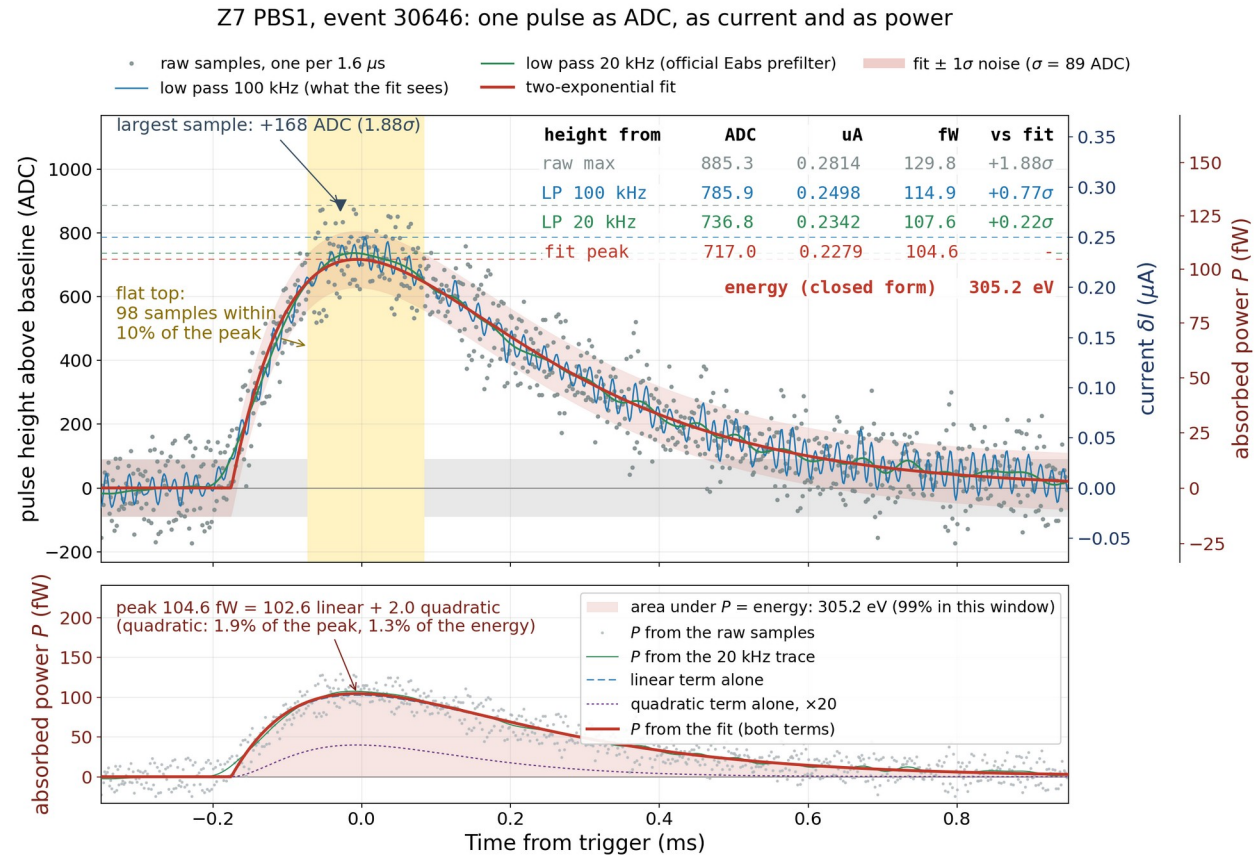
I_0 : TES current at the bias point · R_0 : TES resistance at the bias point · $R_L = R_p + R_{sh}$: load resistance (R_p parasitic, R_{sh} shunt) · δI : change in current, from the trace

The note also gives Method 2: $\delta P = I_0 (R_L - R_0) \cdot \delta I - R_L \cdot (\delta I)^2$ in our sign convention — same linear term, only the quadratic term differs, 1.7 % less energy (PBS1, median of 15 events).

Sources — CDMS wiki, "Collection Efficiency Analysis" (both Methods and the procedure); Method 1 derived in N. Kurinsky, PhD thesis, appendix D (slac.stanford.edu/exp/cdms/ScienceResults/Theses/kurinsky.pdf); Method 2 in S. Watkins, "energy_absorbed_deriv_v3.pdf" on the SLAC confluence page "Internal Documentation: PD2 DM Search v2"; the difference between the two is discussed in "Power_derivation.pdf" on confluence; the TES equations: Irwin & Hilton, Transition-Edge Sensors, Topics in Applied Physics 99 (2005), pp. 74-75 and 87-88.

Example: one pulse

Z7 PBS1, event 30646: the same trace as ADC counts, as current, as power



baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA

$\delta P = I_0(R_L - R_0)\delta I + c_2 R_L (\delta I)^2$ (method 1, $c_2 = 2$), with $I_0 = -18.069 \mu\text{A}$, $R_0 = 44.159 \text{ m}\Omega$, $R_L = 19.240 \text{ m}\Omega$

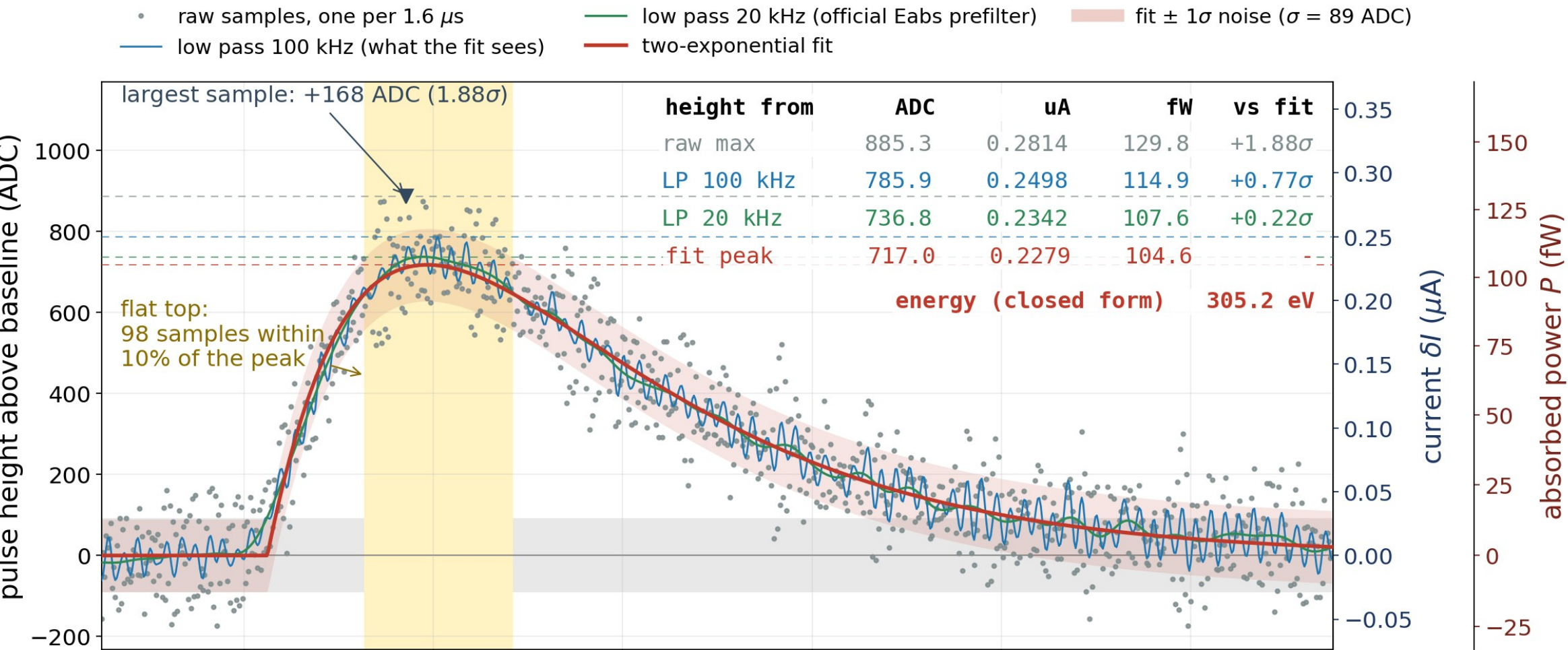
linear coefficient: $I_0(R_L - R_0) = (-18.069 \mu\text{A}) \times (19.240 \text{ m}\Omega - 44.159 \text{ m}\Omega) = 0.450 \mu\text{V}$

quadratic coefficient: $c_2 R_L = 2 \times 19.240 \text{ m}\Omega = 38.48 \text{ m}\Omega = 0.0385 \Omega$

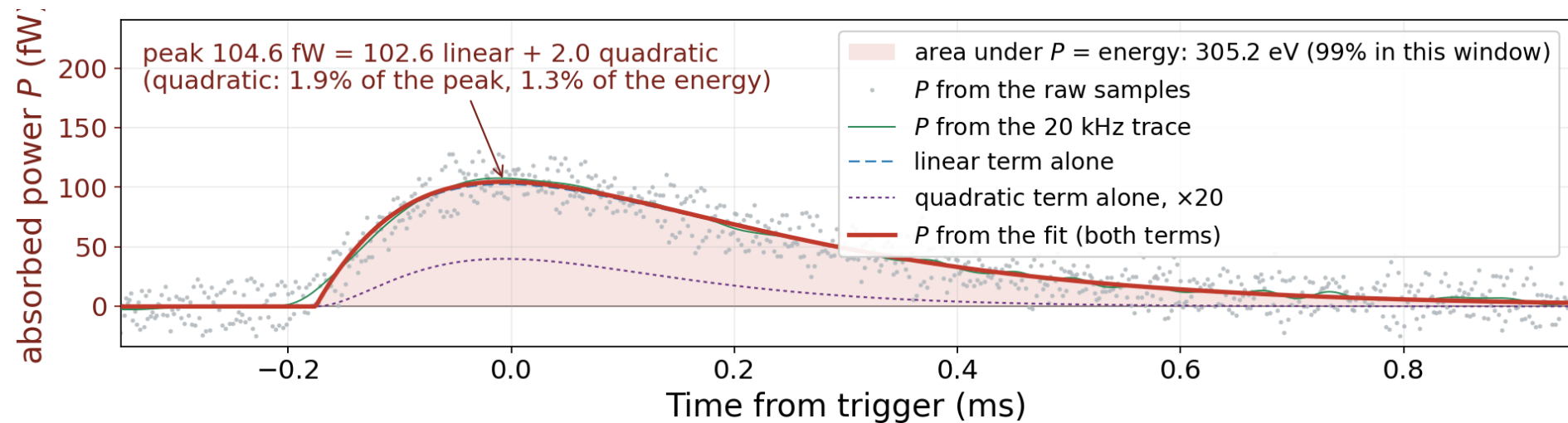
The two-exponential is fitted to the 100 kHz low-passed trace (blue), not to the raw samples; the raw samples are only drawn.

The height: take it from the fit, not from the largest sample

Z7 PBS1, event 30646: one pulse as ADC, as current and as power



The power pulse, and its area is the energy



baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA

$$\delta P = I_0(R_L - R_0)\delta I + c_2 R_L (\delta I)^2 \quad (\text{method1, } c_2 = 2), \quad \text{with } I_0 = -18.069 \mu\text{A}, \quad R_0 = 44.159 \text{ m}\Omega, \quad R_L = 19.240 \text{ m}\Omega$$

linear coefficient: $I_0(R_L - R_0) = (-18.069 \mu\text{A}) \times (19.240 \text{ m}\Omega - 44.159 \text{ m}\Omega) = 0.450 \mu\text{V}$

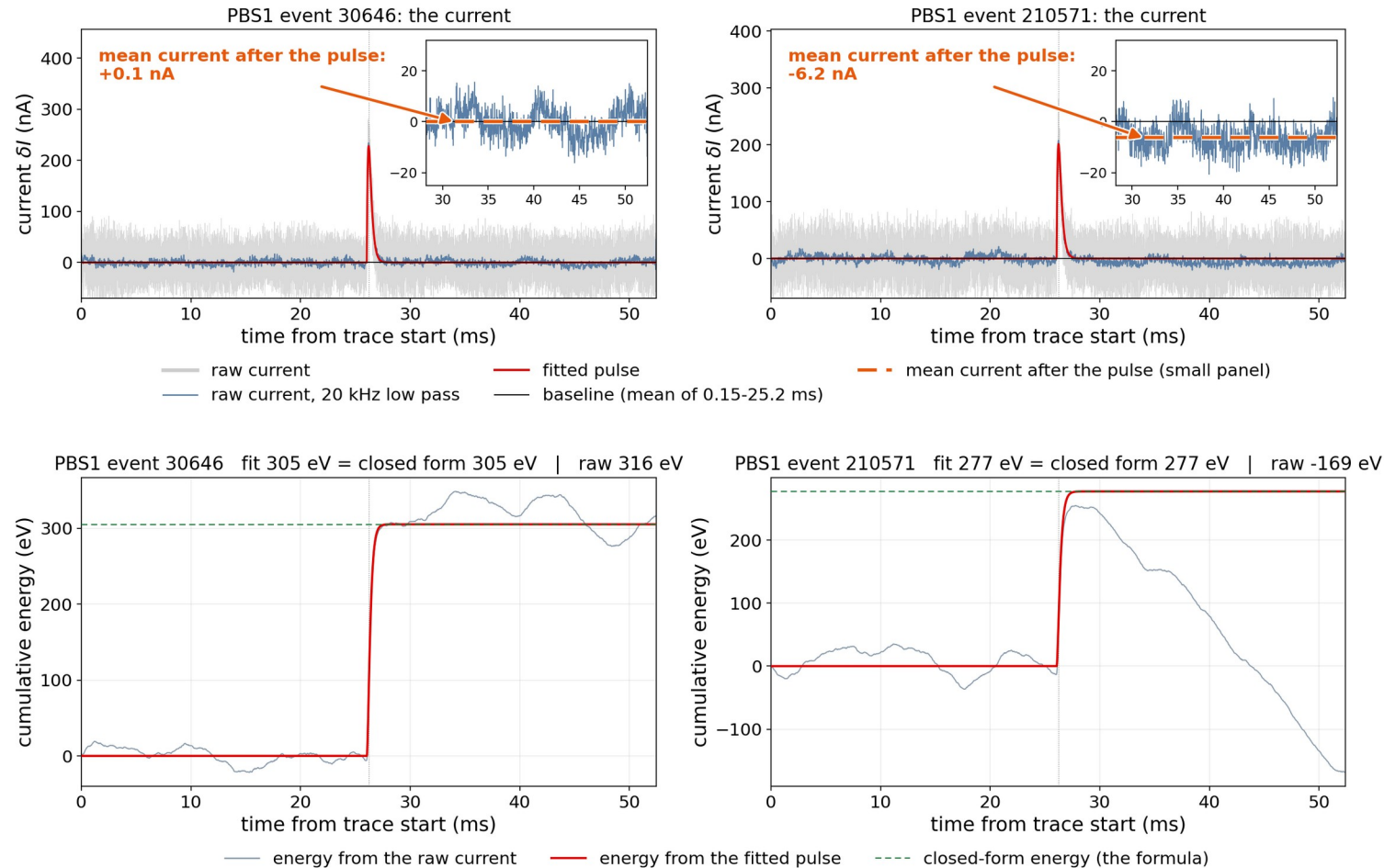
quadratic coefficient: $c_2 R_L = 2 \times 19.240 \text{ m}\Omega = 38.48 \text{ m}\Omega = 0.0385 \Omega$

Red: the fitted current put through the formula. Blue dashed: the linear term alone. Purple: the quadratic term, $\times 20$.

The quadratic term is 1.9 % of the peak — the power pulse has the shape of the current pulse.

The shaded area is the energy this channel absorbed: 305 eV.

Why we integrate the fit, not the raw data

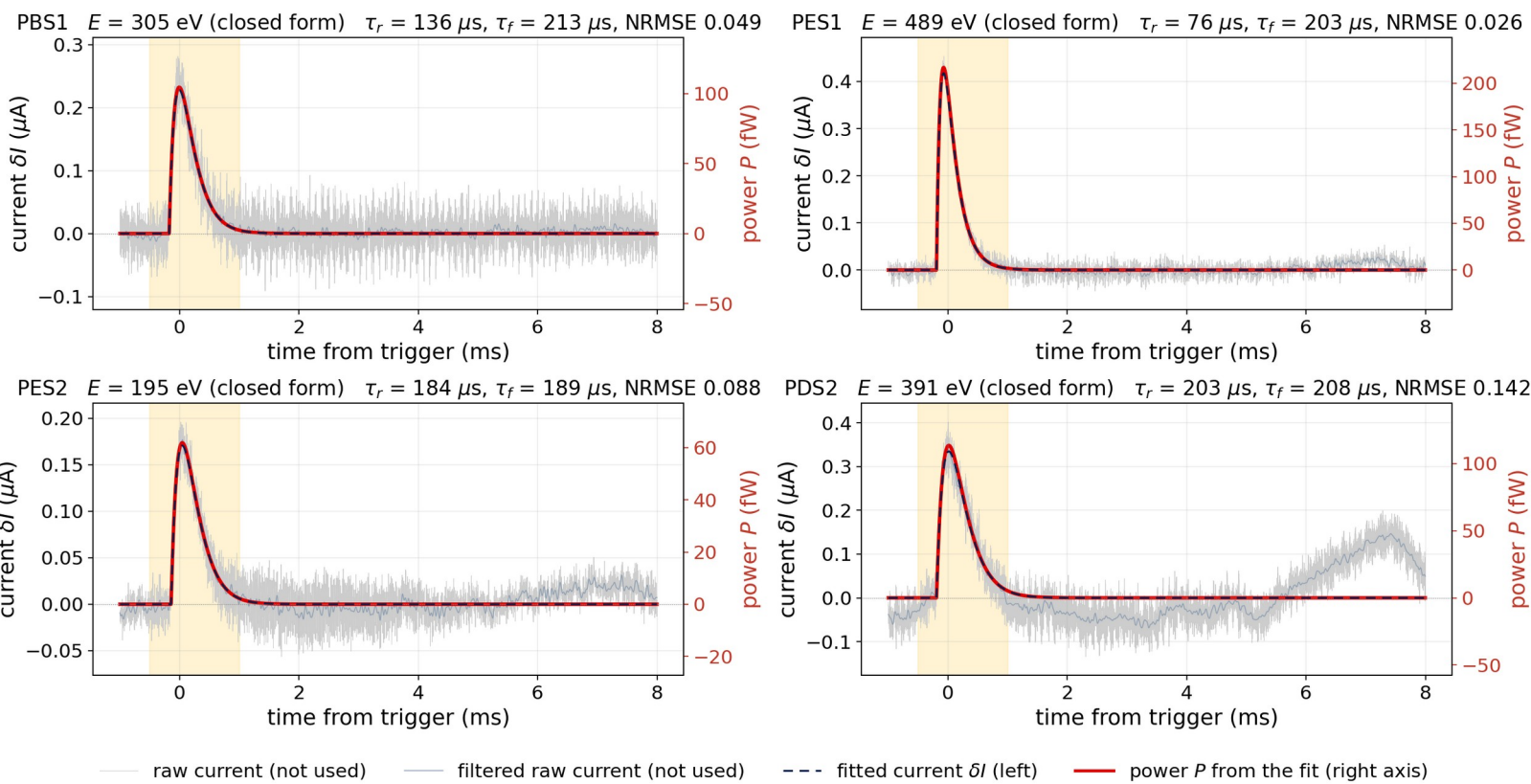


Top: the raw current of the two events; the small panel zooms in on the baseline after the pulse. Bottom: the energy accumulated from the raw current (grey) and from the fit (red).

Right: after the pulse the current sits 6 nA below its baseline — hidden in 28 nA of noise, yet over 24 ms it is -420 eV: raw -169 eV, fit 277 eV.

The same event in its other channels: they add up to 33 %

Z7 event 30646; four of the eleven channels



energy per channel, event 30646

side 1	2062 eV (60 %)
PAS1	303 eV
PBS1	305 eV
PCS1	227 eV
PDS1	378 eV
PES1	489 eV
PFS1	359 eV
side 2	1357 eV (40 %)
PAS2	267 eV
PBS2	261 eV
PCS2	243 eV
PDS2	391 eV
PES2	195 eV

total 3419 eV = 33.0 % of 10.37 keV

PFS2 is not read out on Z7, so side 2 has 5 channels
the channels also differ from each other; see the backup

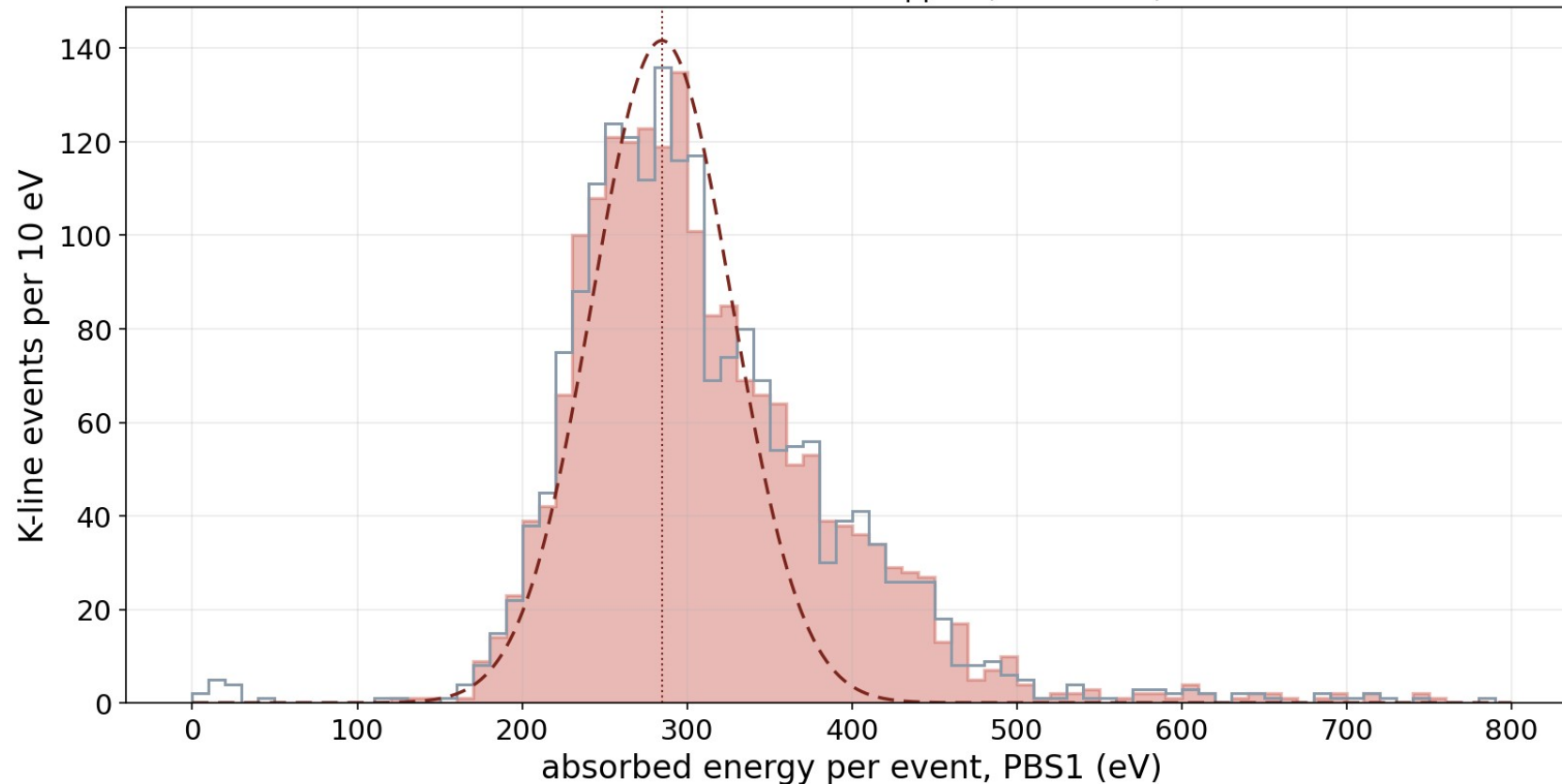
The channels do not share the energy equally: 60 % of it lands on side 1, and PES1 alone has 489 eV against 195 eV in PES2.

Summed over the eleven channels: 3419 eV = 33.0 % of 10.37 keV. PDS2 (bottom right) carries a slow swing.

Now every event: one channel gives a wide, lopsided distribution

PBS1, 1917 K-line events, one fit and one integral each

Z7 PBS1: energy of every K-line event (1917 events, 30 series)
two-exponential fit per event, closed-form power integral (method1)
fits with NRMSE > 0.2 dropped (290 events)



- official-window raw energy: $\mu = 284.2$ eV, $\sigma = 43.1$ eV (15.2%)
- fitted-pulse closed-form energy: $\mu = 284.8$ eV, $\sigma = 42.4$ eV (14.9%)
- Gaussian fit to the core

Red: from the fit. Grey: the official window on the raw trace. Same peak to 0.2 %.

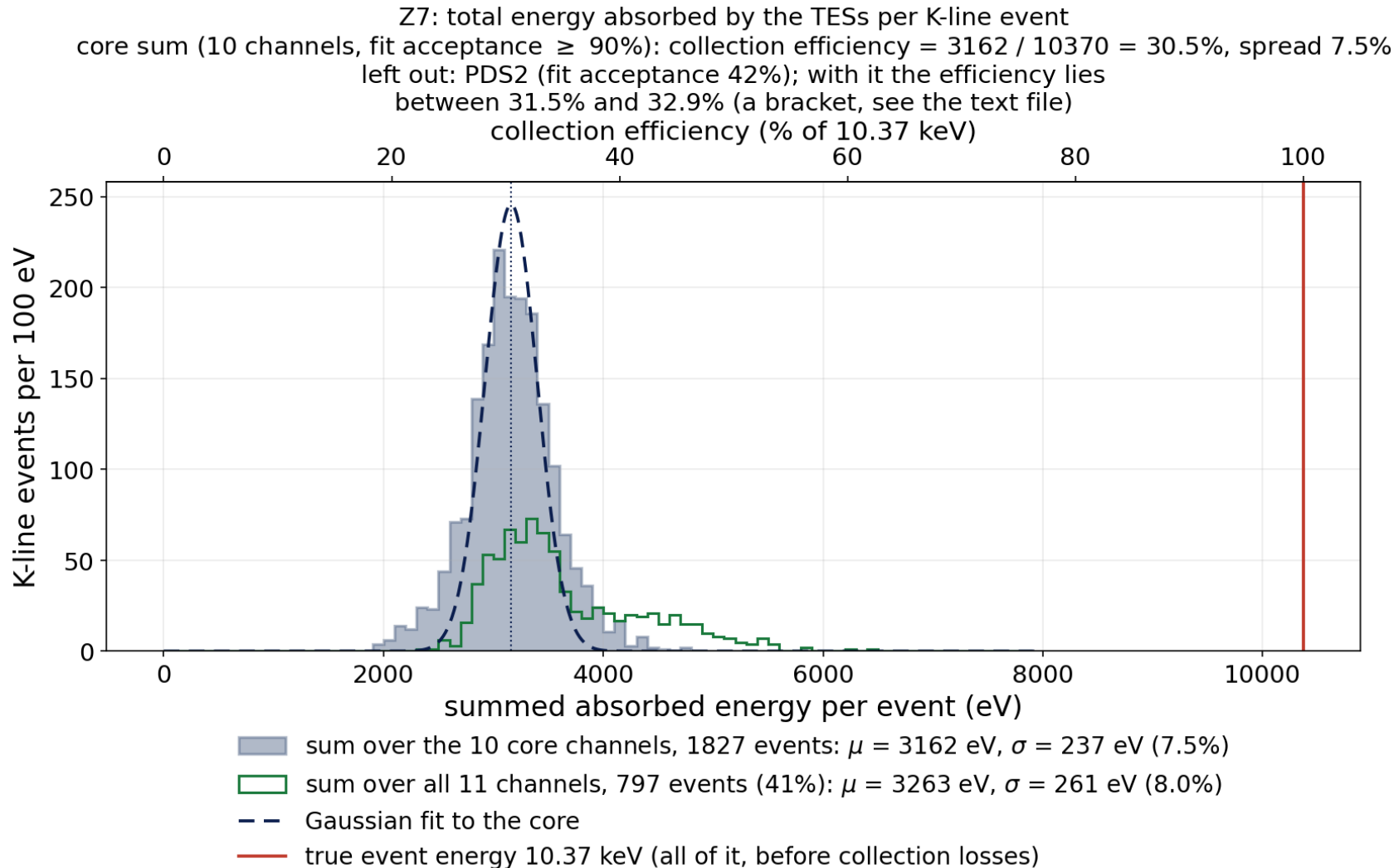
Peak 285 eV, width 15 % — for a line whose energy is fixed.

The tail to the right is the events that happened near this channel.

One channel's width is event position, not noise.

Summed over the channels: 32 ± 1 %

the total energy the TESSs absorbed per K-line event



Ten channels summed, 1827 events: 3162 eV = 30.5 %, and the width drops to 7.5 % — the position effect cancels.

PDS2 fits only 42 % of the events (that slow swing), and those events are not typical; its share can only be bracketed.

Full efficiency 31.5–32.9 %: 32 ± 1 %.

Red line: 10.37 keV — two thirds of the energy never reaches the TESSs.

Summary, and what is still open

A current pulse becomes a power pulse through the TES equations; the area under the power is the energy. The quadratic term is 2 %.

The energy comes from the fitted pulse, integrated from $-\infty$ to $+\infty$: no bias from the noisy maximum, no drift.

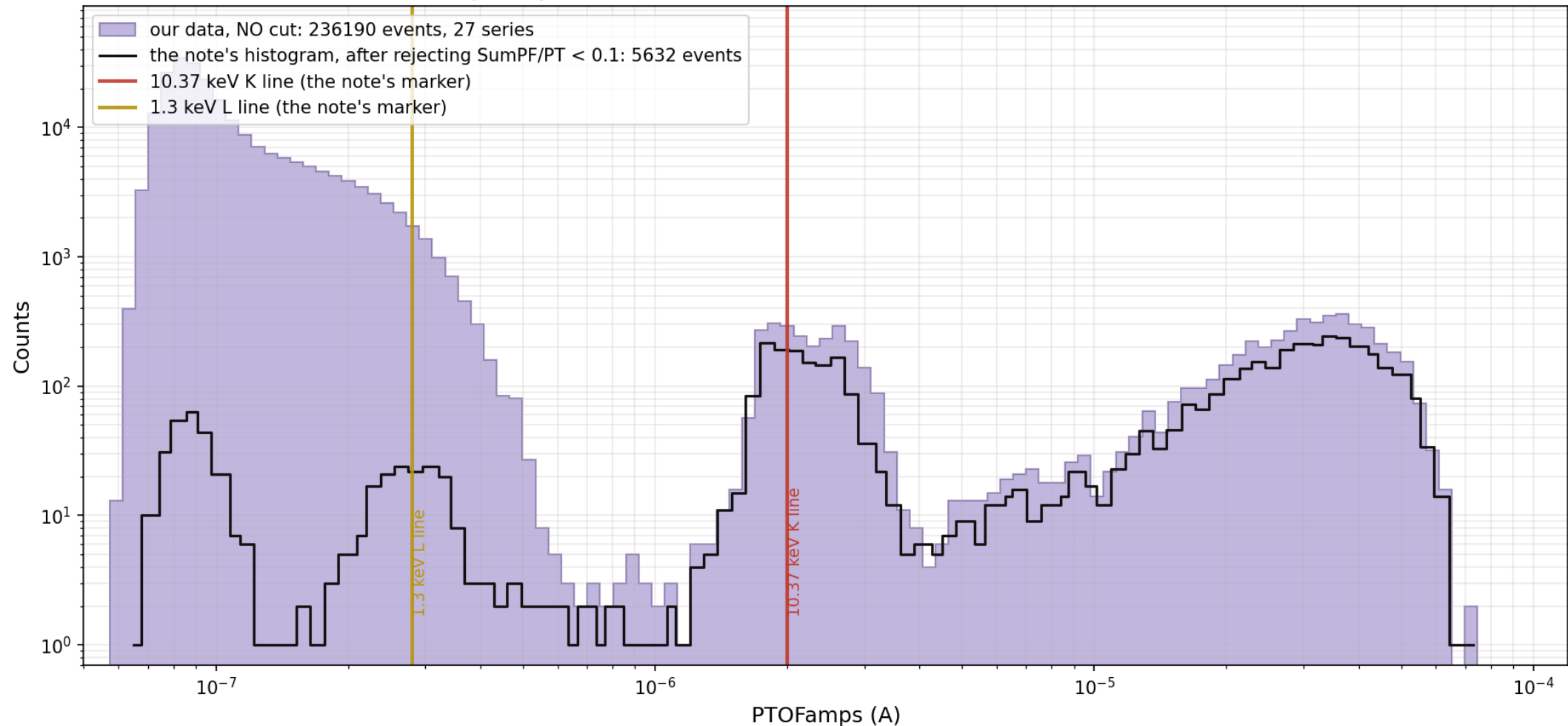
One channel: 15 % wide and lopsided (position). Ten channels summed: 7.5 % and symmetric.

Z7 absorbs 32 ± 1 % of a 10.37 keV event in its TESs. The CDMS note on the R37 CUTE tower reports 26–41 %.

Open: Z7 only · the HV bias of Z7 is not established (0 V assumed) · no dI/dV , so the inductor term is dropped · PDS2 is the ± 1 % · 12.5 % of events fit in no channel and are not in the histograms.

Backup — the events

Z7: summed PTOFamps over 27 series, rebuilt from the Prompt processing with no cut applied
the note's version of the same plot rejects SumPF/PT below 0.1; the difference between the two curves is what that cut removes



Backup — the two Methods of the CDMS note

$\delta P = I_0(R_L - R_0) \cdot \delta I + c_2 \cdot R_L \cdot (\delta I)^2$ with $c_2 = 2$ (Method 1, used here) and $c_2 = -1$ (Method 2)

The two differ only in the quadratic term: 1.7 % in energy (PBS1, median of 15 events) — far below the 15 % single-channel width and the PDS2 bracket.

The inductor term of the full equations needs L and the loop gain from dI/dV , which do not exist for these series on MSI; it is neglected.

The official Eabs (trigger bin 16383, baseline bins 93..15758, 5-pole 20 kHz prefilter, window $-0.5/+1$ ms, same formula) is reproduced to 0.2 %; the grey outlines on the histogram slides are it.

Backup — how the formulas are derived

1. From the current to the absorbed power

heat balance of the TES

$$C \dot{T} = P(t) + P_J - P_{bath}$$

over the whole pulse $\int C \dot{T} dt = 0$; the change in bath heat flow is not included

$$E = \int P dt = - \int \delta P_J dt$$

the circuit; at the bias point $V = I_0(R_L + R_0)$

$$V_{TES} = V - IR_L - L \dot{I}, \quad P_J = IV_{TES}$$

put $I = I_0 + \delta I$ and subtract the value at the bias point

$$\delta P_J = V \delta I - R_L(2I_0 \delta I + (\delta I)^2) - L I \delta \dot{I}$$

integrate: the L term gives $L[I_0 \delta I + (\delta I)^2/2]$, which is 0 at both ends

$$E = \int \delta P dt, \quad \delta P = I_0(R_L - R_0) \delta I + c_2 R_L (\delta I)^2$$

$c_2 = 1$ from this derivation, 2 in Method 1 (used here), -1 in Method 2

2. Energy of the fitted pulse

the fit, with t counted from the pulse start t_0

$$\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$$

the only integral needed

$$\int_0^\infty e^{-t/\tau} dt = \tau$$

linear term

$$\int \delta I dt = A(\tau_f - \tau_r)$$

quadratic term: expand the square, integrate each exponential

$$\int (\delta I)^2 dt = A^2 \left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f \tau_r}{\tau_f + \tau_r} \right]$$

so, with $c_2 = 2$

$$E = I_0(R_L - R_0) A(\tau_f - \tau_r) + 2R_L A^2 \left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f \tau_r}{\tau_f + \tau_r} \right]$$

PBS1, event 30646: $A = 1.39 \mu A$, $\tau_r = 136 \mu s$, $\tau_f = 213 \mu s \Rightarrow E = 305.2 \text{ eV}$

3. Symbols (Z7 PBS1 values) and assumptions

I_b : bias current R_{sh} : shunt resistor, 5.00 m Ω $V = I_b R_{sh}$: bias voltage, $-1.146 \mu V$ R_p : parasitic resistance, 14.24 m Ω

$R_L = R_p + R_{sh}$: load resistance, 19.24 m Ω R_0 : TES resistance at the bias point, 44.16 m Ω $I_0 = V/(R_L + R_0)$: TES current, $-18.07 \mu A$

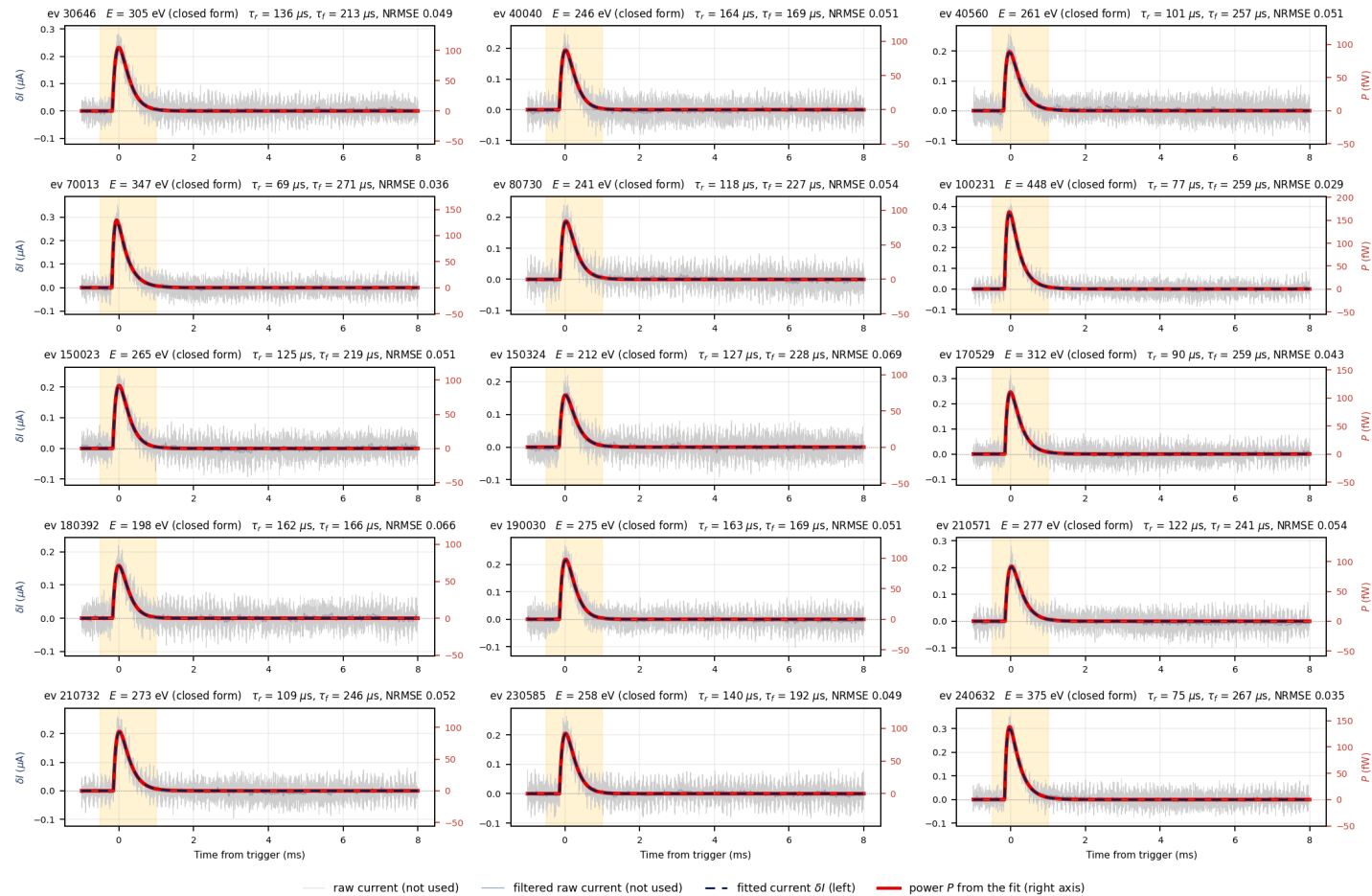
L : inductance (unknown) δI : current change A : amplitude τ_r, τ_f : rise and fall time constants t_0 : pulse start

Assumptions: detector at 0 V (no NTL effect); the heat flow to the bath is not included;

the L term cancels only after integration, so $\delta P(t)$ is not the exact power at each moment; only its integral is used

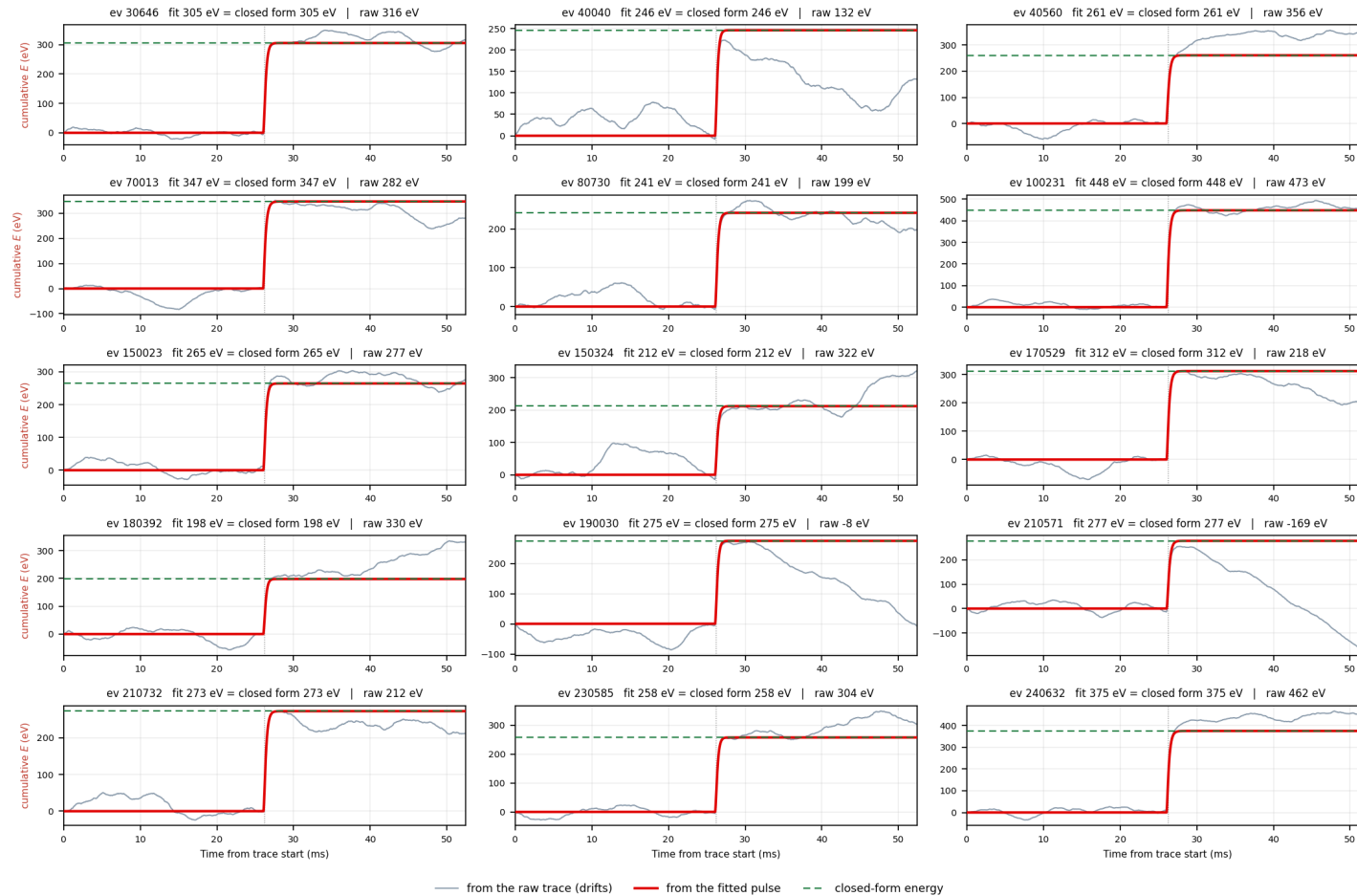
Backup — 15 events, fitted current and power (PBS1)

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
 Z7 24260617_063934, K-line events | method1: $P(t) = 10(R/L-R_0) \cdot dI(t) + 2 \cdot R \cdot L \cdot (dI(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_r) - \exp(-t/\tau_f)]$), 5 free params, pretrigger free in 16383+-3000; the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



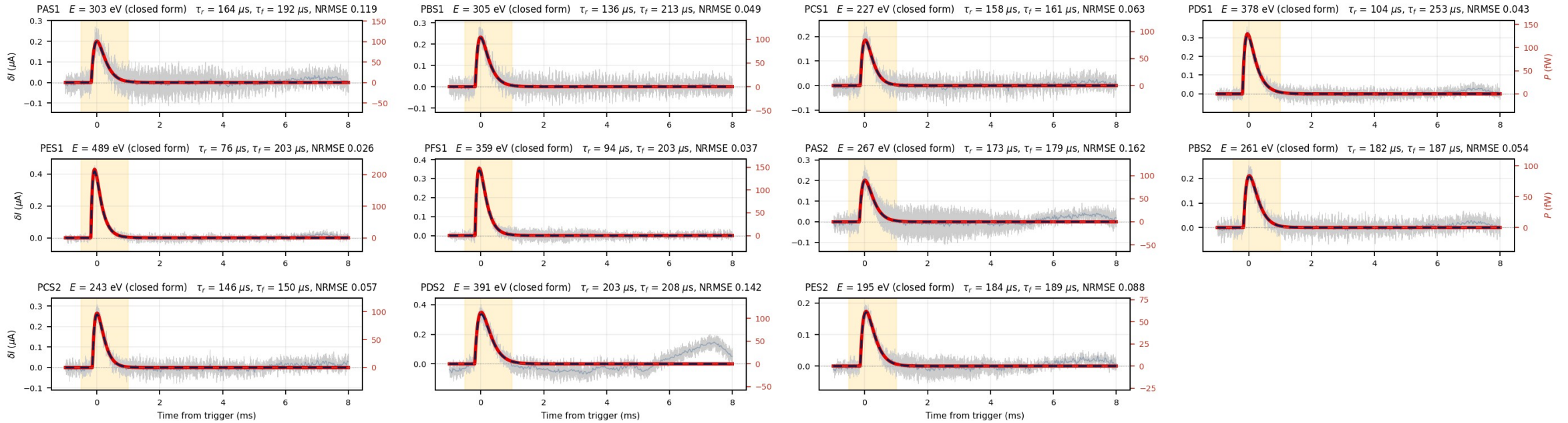
Backup — 15 events, cumulative energy (PBS1)

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace
 Z7 24260617_063934, K-line events | method1: $P(t) = 10(R_L R_0) \cdot d(t) + 2 \cdot R_L \cdot L \cdot (d(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$), 5 free params, pretrigger free in 16383+-3000; the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — event 30646, all channels

Z7 event 30646, all channels: fitted current and power
sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_{L-R0}) * dl(t) + 2 * R_{L-R0} * L * (dl(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$, 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — energy distribution of every channel

Z7: energy of every K-line event, one panel per channel (30 series)

each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic

the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3710 eV = 35.8% of the 10.37 keV K line)

